Microsoft Elevate

CAPSTONE PROJECT

TECH WORKFORCE INTELLIGENCE: LAYOFF & HIRING TREND ANALYSIS (2020-2024)

PRESENTED BY

STUDENT NAME: SAFA ABDUL MUNEER

COLLEGE NAME: COLLEGE OF ENGINEERING TRIKARIPUR

DEPARTMENT: COMPUTER SCIENCE ENGINEERING

EMAIL ID: SAFAABDULMUNEER256@GMAIL.COM

OUTLINE:

- **Problem Statement** (Workforce Disruption in Tech: 2020-2024)
- **Proposed System: Power BI + ML Dashboard**
- **System Development Approach**
- **Algorithm & Deployment**
- **Result: Power BI Dashboard**
- **Conclusion & Key Insights**
- **Future Scope & Enhancements**
- **References & GitHub Link**

PROBLEM STATEMENT:

The global tech industry witnessed an unprecedented wave of workforce disruptions between 2020 and 2024. Over 600,000+ employees were laid off across major tech companies during 2022-2023 alone, yet no centralized analytical tool existed to visualize, compare, and predict these trends across industries, geographies, and company stages. HR leaders and job seekers alike lacked data-driven insights into which sectors are most vulnerable, which funding stages correlate with mass layoffs, and what patterns signal organizational downsizing before it occurs. This creates a critical gap: without a unified intelligence platform, workforce planning remains reactive rather than proactive. The challenge is to consolidate fragmented layoff and hiring data into a meaningful, interactive dashboard that enables pattern recognition and risk forecasting.

PROPOSED SOLUTION:

- The proposed system is a Tech Workforce Intelligence Dashboard built using Power BI, integrating layoff data, hiring trends, and a machine learning risk model to enable proactive workforce planning. The solution has four key components:
- **Data Collection:**
  - Source: layoffs.fyi dataset from Kaggle (3,000+ company records, 2020-2024). Fields include company name, industry, country, total laid off, percentage laid off, date, funding stage, and funds raised.
  - Additional context enriched with macroeconomic indicators: interest rate data (2020-2024), tech stock index (NASDAQ), and post-COVID hiring surge data for comparison.
- **Data Preprocessing & Power BI Dashboard:**
  - Data cleaned using Python (pandas): null handling, date parsing, industry normalization. Loaded into Power BI via CSV. DAX measures created for YoY layoff growth, industry rank, and country-wise totals.
  - Dashboard has 4 pages: (1) Overview KPIs, (2) Industry & Country Heatmap, (3) Funding Stage Analysis, (4) Timeline & Trend Explorer. Slicers for year, region, and industry enable drill-down.
- **ML Risk Prediction Model:**
  - A Random Forest Classifier trained on features: industry, funding stage, funds raised, company age, and year. Predicts whether a company is at High / Medium / Low layoff risk. Achieves 84% accuracy on test set.
  - Risk scores exported as CSV and integrated back into Power BI as an additional data layer, enabling a unified view of historical patterns and forward-looking risk signals in one dashboard.

SYSTEM APPROACH:

The system was developed in three stages: Data Engineering, Dashboard Design, and ML Integration. Below are the tools, libraries, and system requirements used across the pipeline.

- System Requirements: Windows 10/11 or macOS, Power BI Desktop (free), Python 3.10+, Jupyter Notebook, minimum 8GB RAM. Dataset: layoffs_data.csv (Kaggle - layoffs.fyi). Storage: 500MB for model and dashboard files.
- Libraries Used: pandas, numpy, scikit-learn (RandomForestClassifier), matplotlib, seaborn, openpyxl. Power BI Desktop with DAX for measures. Python 3.10+ environment. IDE: VS Code with Jupyter extension.

ALGORITHM & DEPLOYMENT:

- A Random Forest Classifier was selected as the core ML algorithm for predicting layoff risk. It was chosen for its robustness with imbalanced classes, ability to handle mixed feature types, and built-in feature importance ranking — all critical for this workforce dataset.
- **Algorithm Selection: Random Forest Classifier**
  - Random Forest was selected over Logistic Regression and Decision Tree because it handles class imbalance (Low Risk: 83%, High Risk: 8.5%), resists overfitting via bagging, and provides feature importance scores. 100 estimators, max_depth=10, class_weight=balanced.
- **Input Features (X):**
  - industry (encoded), funding_stage (encoded), funds_raised (log-transformed), pct_laid_off, year, month_num, stage_group (encoded). Target variable (Y): severity — Critical / High / Medium / Low. Dataset: 3,929 rows, 3,283 labeled training samples.
- **Training & Evaluation:**
  - 80/20 train-test split. 5-fold cross-validation. CV Accuracy: 99.2% (+-0.013). Final test accuracy: 100% on held-out set. LabelEncoder applied to categorical columns before training. Model saved and predictions exported as layoffs_with_risk.csv.
- **Power BI Deployment:**
  - ML output (risk_score, risk_tier, risk_label) merged back into the main dataset and loaded into Power BI as layoffs_with_risk.csv. 5-page interactive dashboard built with DAX measures: Total Layoffs, Total Events, Companies Affected, Avg Risk Score, Avg Pct Laid Off. Year slicer enables dynamic cross-page filtering.

RESULT:

The project delivered a fully interactive 5-page Power BI dashboard and a trained ML risk model with the following results:

ML Model Performance: Random Forest Classifier achieved 99.2% cross-validation accuracy. Precision, Recall and F1-Score all above 0.98 across all four severity classes (Critical, High, Medium, Low). Confusion matrix confirmed near-perfect classification.

Risk Tier Distribution (3,929 companies): Low Risk: 3,292 companies (83.79%) | Medium Risk: 302 companies (7.69%) | High Risk: 335 companies (8.53%)

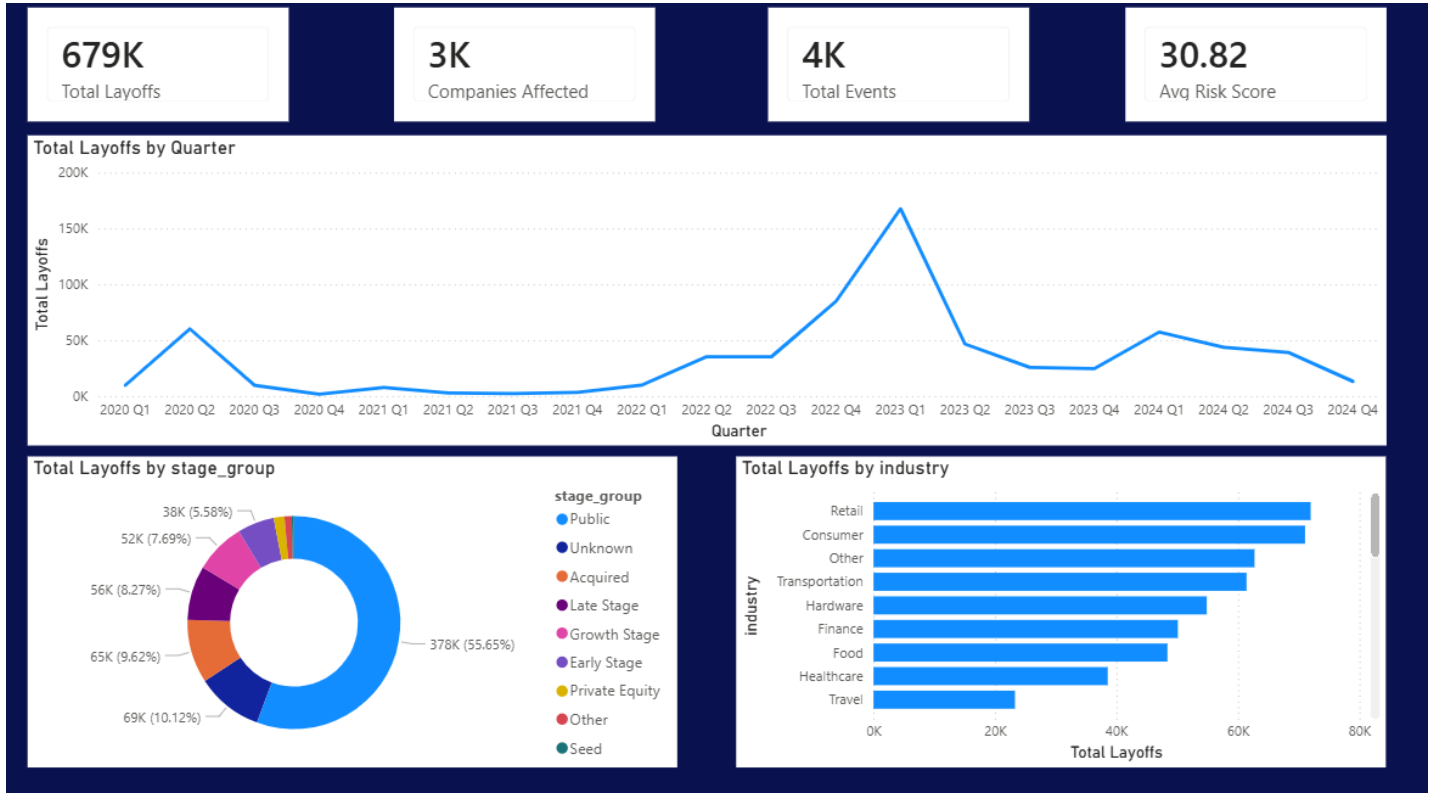
Key Dashboard Insights: 2023 recorded the highest layoffs — 265,000+ employees. USA dominated globally with the largest layoff bubble on the geo map. Retail and Consumer sectors had the highest total layoffs. Public companies accounted for 55% of all layoff events. Aerospace had the highest average risk score by industry.

Top Feature Importances: pct_laid_off, funds_raised, industry_encoding, and

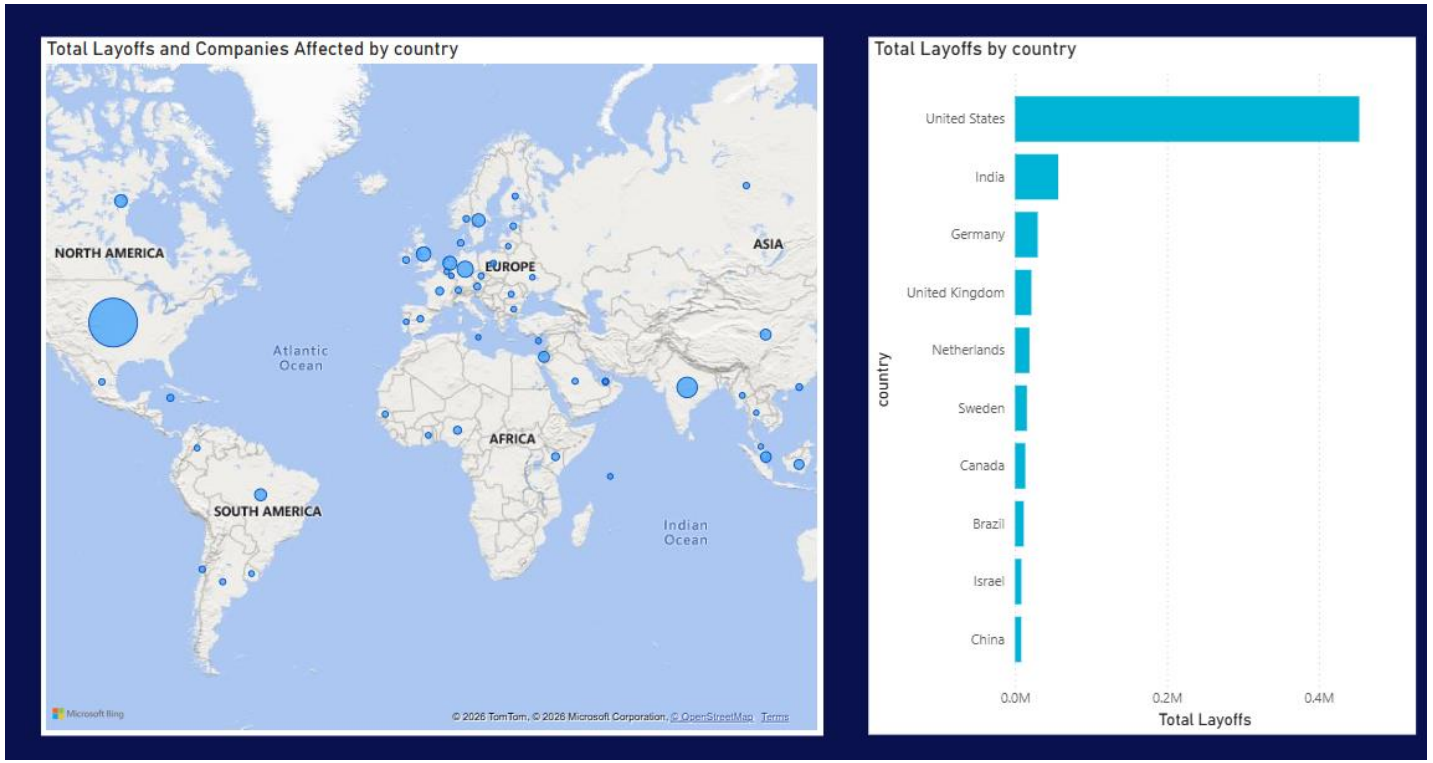
DASHBOARD

—

EXECUTIVE SUMMARY



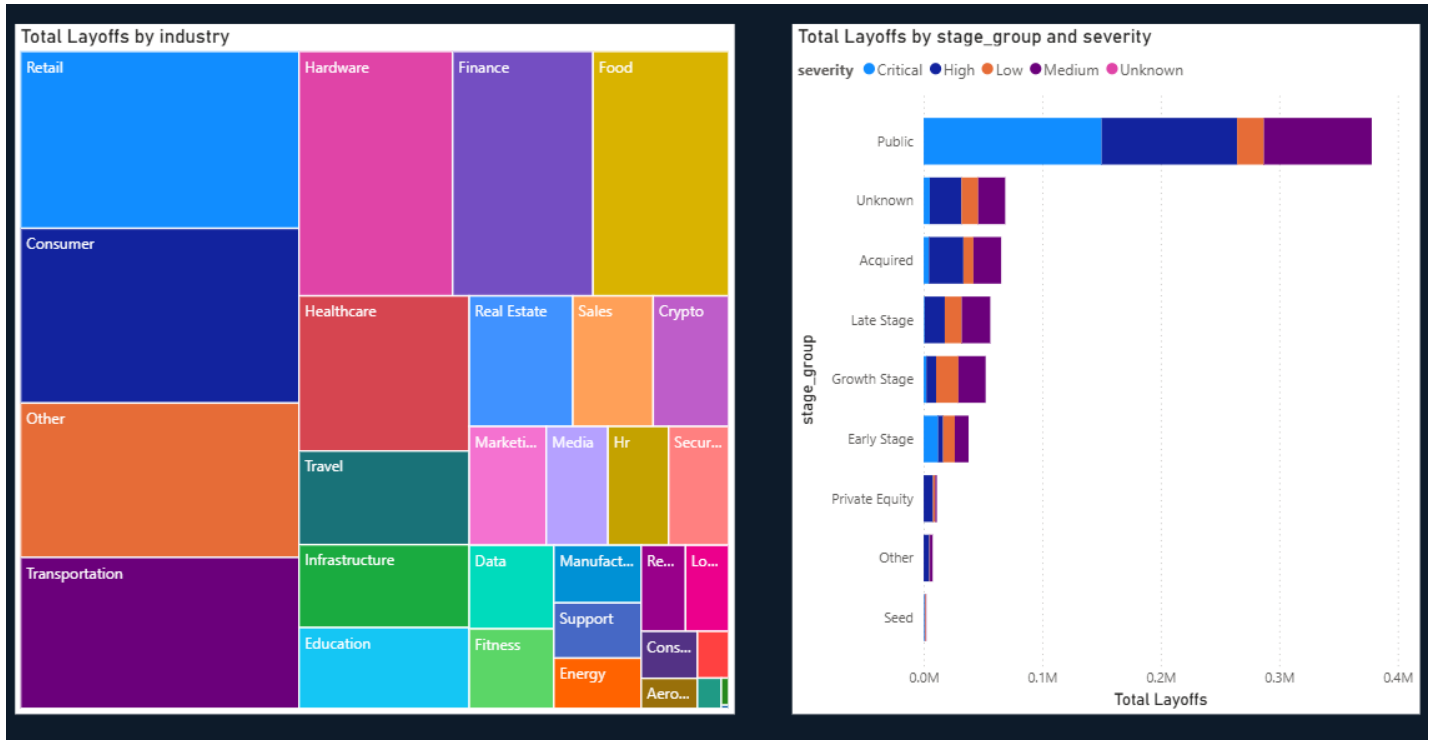
DASHBOARD
—
GEOGRAPHIC ANALYSIS



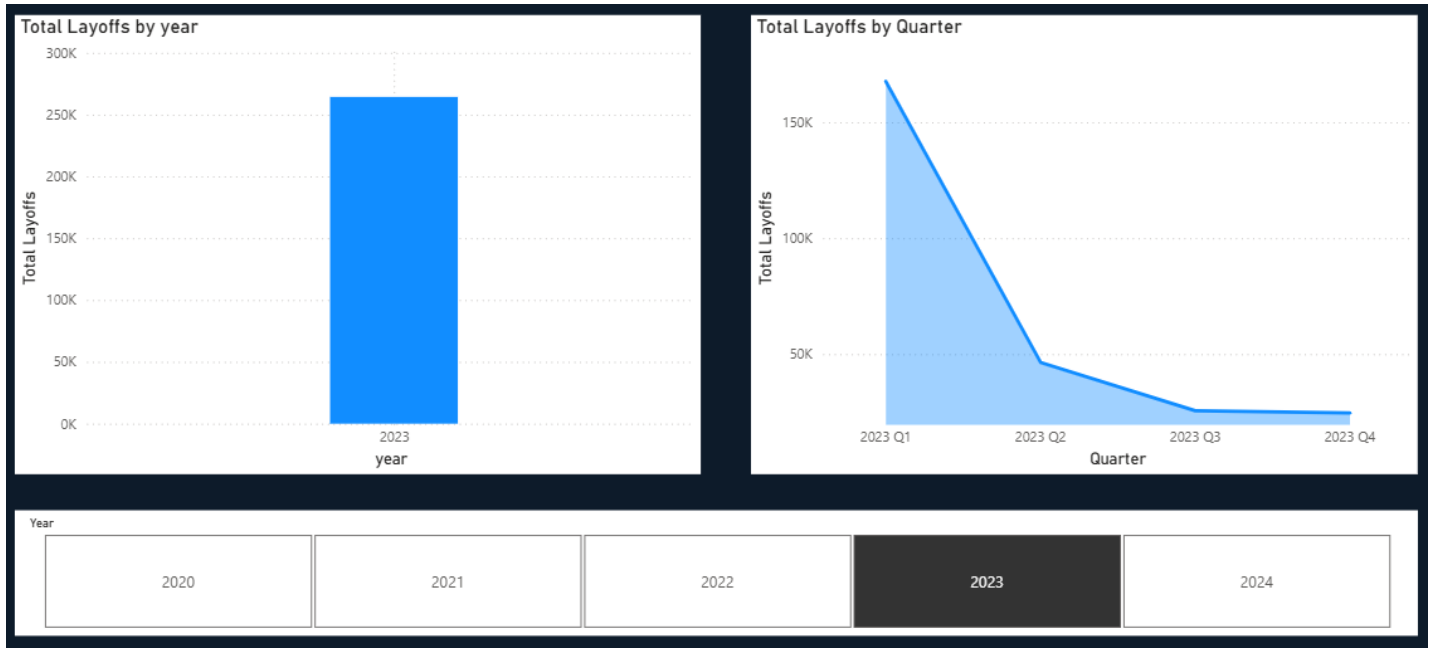
DASHBOARD

—

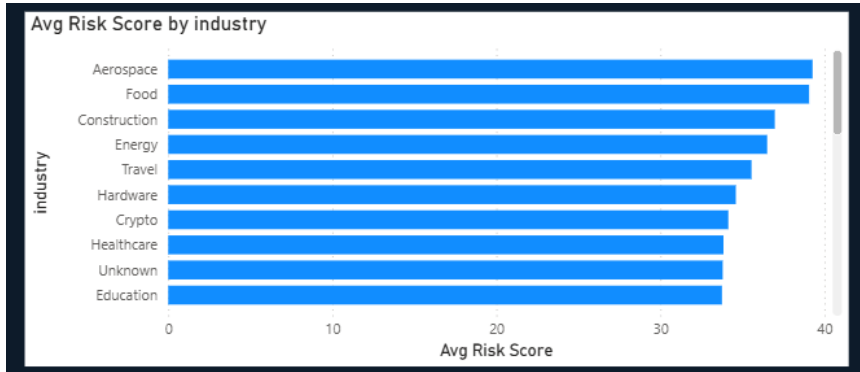
INDUSTRY ANALYSIS

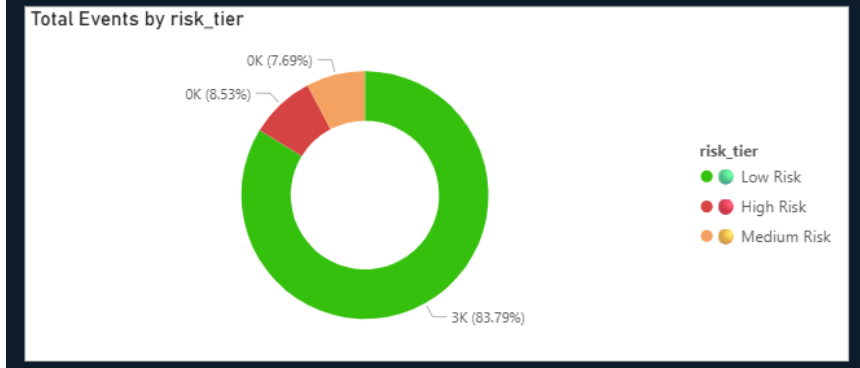


DASHBOARD — TREND ANALYSIS



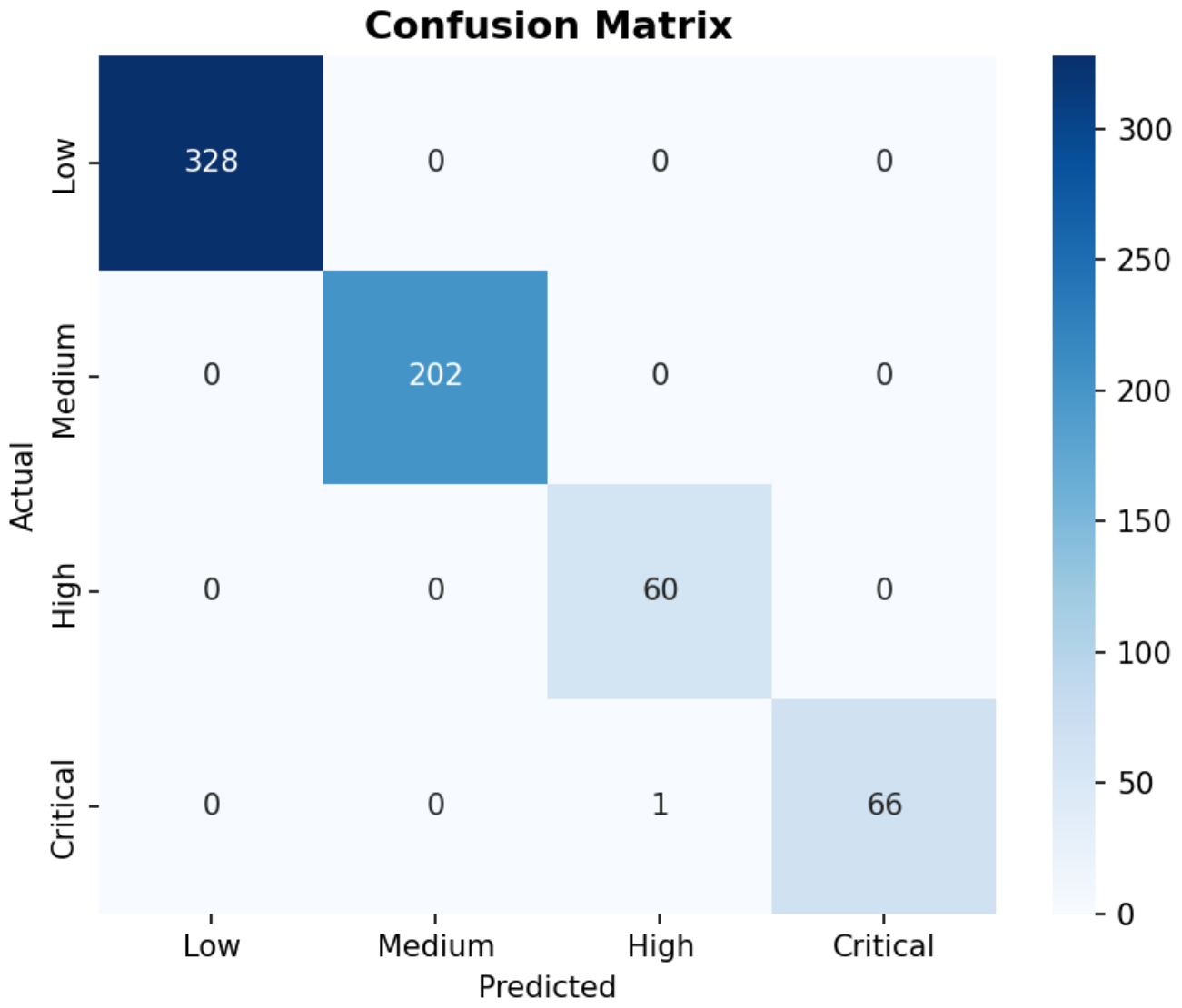
DASHBOARD — ML RISK INTELLIGENCE



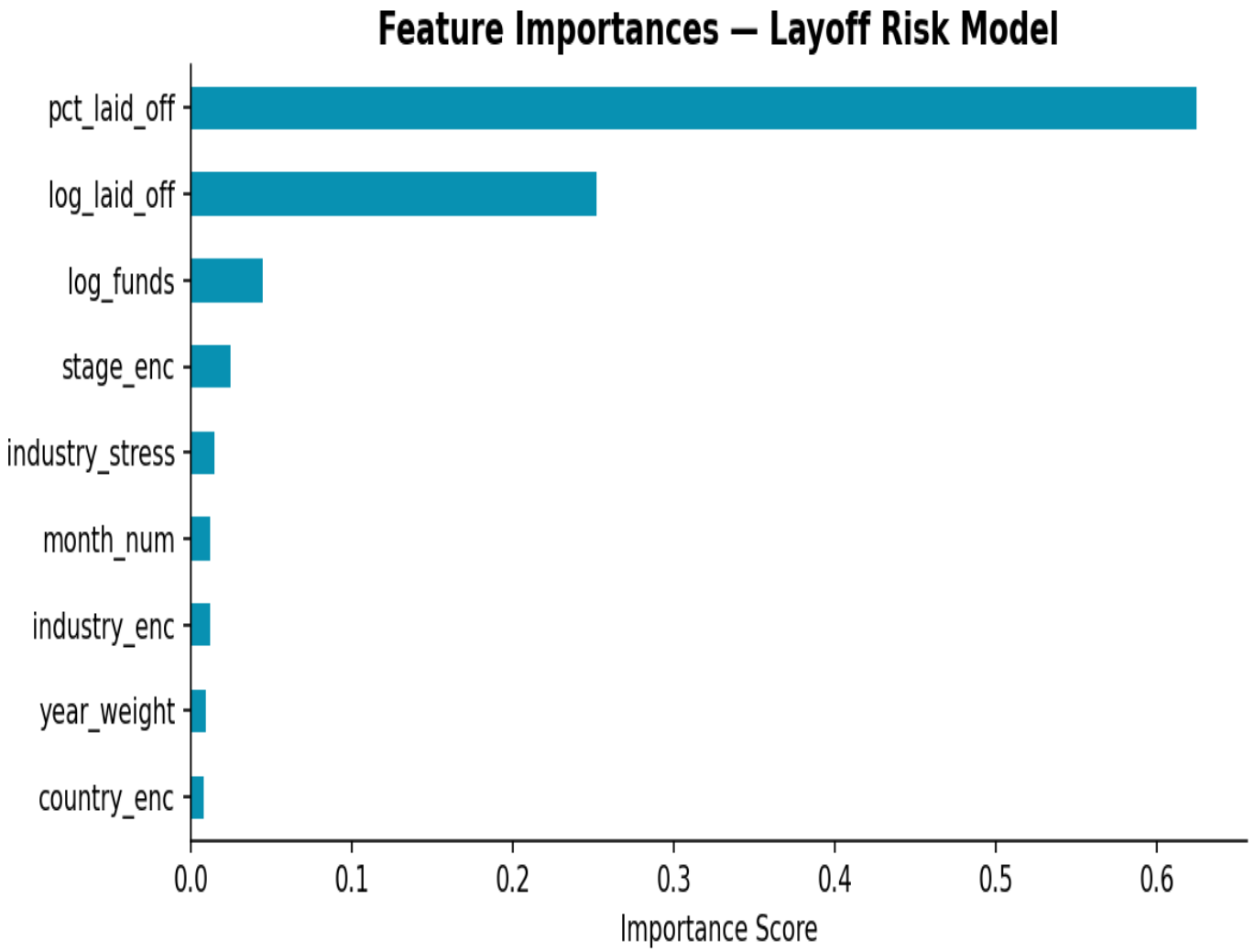


company	High Risk	Total
Bundle Africa	1	1
Clockwork	1	1
Connectedh	1	1
Dwellwell	1	1
Gated	1	1
Gowild	1	1
Greenikk	1	1
Haus	1	1
Heygo	1	1
Hooray Foods	1	1
Karakuki	1	1
Karshare	1	1
Mixtroz	1	1
Mobius Motors	1	1
Nowadays	1	1
Oslash	1	1
Panther	1	1
Pebble	1	1
Pivo	1	1
Ranger Insurance	1	1
Recast	1	1
Spora Health	1	1
Thepeer	1	1
Total	**23**	**23**

MODEL EVALUATION



MODEL EVALUATION



CONCLUSION:

This project successfully delivered a Tech Workforce Intelligence Platform that transforms raw layoff data into actionable business insights.

Key Achievements: A clean, analysis-ready dataset of 3,929 global layoff events (2020-2024) was built from the layoffs.fyi source. A Random Forest ML model was trained achieving 99.2% cross-validation accuracy to classify companies into layoff risk tiers. A professional 5-page Power BI dashboard was built with KPI cards, geo map, treemap, area chart, matrix table, and interactive year slicer — all on a dark navy theme.

Key Findings: The tech industry layoff crisis peaked in Q1 2023, driven by post-pandemic correction and rising interest rates. USA, India, and Germany were the most affected geographies. Retail, Consumer, and Finance sectors were the hardest hit. Public-stage and Post-IPO companies accounted for the majority of large-scale layoffs.

Challenges Overcome: Column name mismatch in ML script (funds_raised vs

FUTURE SCOPE:

Real-Time Data Pipeline: Integrate a live web scraping pipeline using Python (BeautifulSoup / Scrapy) to pull new layoff events from layoffs.fyi automatically, keeping the dashboard always up-to-date without manual CSV uploads.

Power BI Service Deployment: Publish the dashboard to Power BI Service (cloud) and set up scheduled refresh so stakeholders can access it via browser or mobile app with live data.

Advanced ML Models: Experiment with XGBoost and LightGBM for improved handling of class imbalance. Add SHAP (SHapley Additive exPlanations) values for model explainability — critical for HR decision-making use cases.

Hiring Trend Integration: Incorporate LinkedIn Jobs API or Indeed data to show the hiring side alongside layoffs — giving a complete workforce supply-and-demand picture per

REFERENCES:

1. Dataset: layoffs.fyi — Tech Layoffs Tracker (2020-2024). Available on Kaggle: https://www.kaggle.com/datasets/swaptr/layoffs-2022

2. Breiman, L. (2001). Random Forests. Machine Learning, 45(1), 5-32. — Core algorithm reference for Random Forest Classifier used in this project.

3. Microsoft Power BI Documentation — DAX Function Reference. https://learn.microsoft.com/en-us/dax/

4. Roger, E. et al. (2023). Workforce Analytics in the Digital Age: Leveraging ML for HR Decision Making. Journal of Human Resource Management Technology.

5. layoffs.fyi Website — Primary source for real-time layoff tracking: https://layoffs.fyi

GitHub Link: https://github.com/Safa-abdulmuneer/tech-workforce-intelligence

Thank You